Replica twist entropy fails strong monotonicity under fermionic Gaussian measurements

Xiaojian Zeng*
PBC School of Finance, Tsinghua University, Beijing 100190, China
(Dated: October 14, 2026)

Replica twist entropy is a two-copy diagnostic of pure-state fermionic non-Gaussianity with Gaussian invariance and product additivity. We show that it does not satisfy strong monotonicity under recorded fermionic Gaussian measurements. A coherent family with two occupation flags has definite even parity and admits a closed replica overlap for every integer number of four-mode resource blocks and every interior rotation angle. Measuring one flag retains two pure branches of definite parity. At the reflection-symmetric angle, an explicit 34-mode input with outcome probabilities 3/4 and 1/4 increases the entropy on average by exactly $\ln(25836385/25165824) > 0$. For each fixed interior angle and nonzero branch probability, the input entropy stays bounded while the mean output entropy grows linearly with the number of blocks. This is an operational limitation of the finite-angle replica diagnostic, without changing its established diagnostic properties. The coherent-dilution mechanism is known in stabilizer resource theory; the result here supplies its parity-preserving fermionic realization and exact graded-copy calculation for the replica twist observable.

I. INTRODUCTION

Fermionic non-Gaussianity supplies resources beyond free-fermion dynamics. Several ways to quantify it use covariance matrices, convolution, or two-copy invariants [1–5]. These constructions address different operational questions. In particular, a faithful diagnostic that is invariant under Gaussian unitaries and additive on products need not decrease on average under Gaussian measurements.

Sun and Zhang introduced replica twist entropy as a finite-angle two-copy probe of pure-state non-Gaussianity and applied it to Sachdev–Ye–Kitaev dynamics [6]. Its finite-angle overlap probes correlations beyond the covariance matrix. Their defining work establishes Gaussian invariance, additivity, and diagnostic properties; it does not assert the strong-monotonicity inequality examined here.

The distinction between logarithmic resource diagnostics and strong monotones has a close precedent in stabilizer theory. Haug and Piroli showed that stabilizer entropies fail strong monotonicity using coherent dilution [7]. Leone and Bittel subsequently distinguished logarithmic stabilizer monotones from their strongly monotone linear counterparts and gave a flagged construction [8]. We use this known idea in the fermionic setting. Two flags enforce fermion-parity superselection, and the rotation amplitude must be computed with anticommuting copies. The contribution is an explicit family for the actual replica twist functional, its closed all-angle expression, a rational finite-mode counterexample, and unbounded average failure at every fixed interior angle. The four-mode block used below is itself a qubit stabilizer state, so the calculation is not an identification of stabilizer and fermionic resource measures.

* xiaojianzeng@foxmail.com

II. OBSERVABLE AND OPERATIONAL QUESTION

Let $\mathcal{H}_N$ be the Fock space of N complex fermionic modes. Hermitian Majoranas γ_j, $j = 1, \ldots, 2N$, obey $\{\gamma_j, \gamma_k\} = 2\delta_{jk}$. All input vectors are normalized parity eigenstates, $\Pi_N|\psi\rangle = \epsilon|\psi\rangle$, with $\epsilon = \pm 1$. Coherence between the two parity sectors is excluded. On two graded copies, a register-first ordinary tensor representation is

$$\gamma_j^{(1)} = \gamma_j \otimes I, \qquad \gamma_j^{(2)} = \Pi_N \otimes \gamma_j. \tag{1}$$

These Majoranas anticommute across copies as well as within each copy. The replica rotation and its overlap are

$$U_{\mathrm{R}}(\alpha) = \exp\left[\frac{\alpha}{2}\sum_{j=1}^{2N}\gamma_j^{(1)}\gamma_j^{(2)}\right], \tag{2}$$

$$F_N(\psi;\alpha) = \langle\psi|^{\otimes 2} U_{\mathrm{R}}(\alpha)|\psi\rangle^{\otimes 2}, \tag{3}$$

$$P_N(\psi;\alpha) = -\ln F_N(\psi;\alpha), \qquad 0 < \alpha < \frac{\pi}{2}. \tag{4}$$

Angles and entropies are dimensionless, and logarithms are natural. The positive Majorana-string expansion recalled in Appendix A gives $0 < F_N \le 1$, so no logarithm-branch convention is needed. The factor $1/2$ in Eq. (2) is part of the source convention; replacing the graded second copy by $I \otimes \gamma_j$ changes the model.

Pure Gaussian states have $P = 0$. Gaussian-unitary invariance and additivity for fixed-parity product states are established properties [6]. A strong monotone would further obey

$$P_N(\psi;\alpha) \;\ge\; \sum_{a:p_a>0} p_a P_{N_a}(\phi_a;\alpha) \tag{5}$$

under Gaussian protocols with recorded pure outcomes. Here the allowed primitives are even quadratic-Hamiltonian unitaries, pure Gaussian ancillas of definite parity, occupation measurements, and conditional

Gaussian operations. A removed mode is refined into a recorded occupation measurement; the compared output branches are pure. No formula for a general mixed output is assumed.

It suffices to violate Eq. (5) with one allowed occupation measurement, retaining both outcomes. The original question fixes $\alpha = \pi/4$; the same family also permits an exact answer at every fixed interior angle.

III. A PARITY-PRESERVING FAMILY AND ITS OVERLAP

Define the locally even four-mode vectors

$$|V\rangle = |0000\rangle, \qquad |C\rangle = \frac{|0000\rangle + |1111\rangle}{\sqrt{2}}. \tag{6}$$

For an integer $m \geq 1$ and $0 < q < 1$, set $p = 1 - q$ and

$$|A_m\rangle = |00\rangle|V\rangle^{\otimes m}, \qquad |B_m\rangle = |11\rangle|C\rangle^{\otimes m}, \tag{7}$$

$$|\psi_{m,q}\rangle = \sqrt{p}\,|A_m\rangle + \sqrt{q}\,|B_m\rangle, \qquad N = 4m + 2. \tag{8}$$

The two flag modes precede the resource blocks in Fock order. The flags make A_m and B_m orthogonal; every supported word has even occupation. Thus the input is normalized and has definite even total parity. Its preparation by free operations is not assumed. The second flag ensures that the superposition respects parity superselection.

Measuring the occupation of the first flag gives

$$\begin{aligned}(p_0, |\phi_0\rangle) &= (p, |0\rangle|V\rangle^{\otimes m}),\\ (p_1, |\phi_1\rangle) &= (q, |1\rangle|C\rangle^{\otimes m}).\end{aligned} \tag{9}$$

Both outcomes are recorded, both outputs are normalized pure states, and their remaining parities are respectively even and odd. There is no ancillary resource or additional gate in this operation.

A. Graded rotation

Write $f_\ell^{(r)} = (\gamma_{2\ell-1}^{(r)} + i\gamma_{2\ell}^{(r)})/2$. Using the cross-copy CAR in Eq. (1),

$$U_{\mathrm{R}}(\alpha) = \exp\left[\alpha \sum_{\ell=1}^{N} \left(f_\ell^{(1)\dagger} f_\ell^{(2)} - f_\ell^{(2)\dagger} f_\ell^{(1)}\right)\right]. \tag{10}$$

It fixes the joint vacuum. With $u = \cos\alpha$ and $v = \sin\alpha$, its action on creation operators is

$$\begin{aligned}U_{\mathrm{R}} f_\ell^{(1)\dagger} U_{\mathrm{R}}^\dagger &= u f_\ell^{(1)\dagger} - v f_\ell^{(2)\dagger},\\ U_{\mathrm{R}} f_\ell^{(2)\dagger} U_{\mathrm{R}}^\dagger &= v f_\ell^{(1)\dagger} + u f_\ell^{(2)\dagger}.\end{aligned} \tag{11}$$

The corresponding one-particle matrix is

$$R = \begin{pmatrix} uI & vI \\ -vI & uI \end{pmatrix}. \tag{12}$$

For ordered occupied row and column lists J, K, the Fock-space matrix element is the occupied minor $\det R[J, K]$. This follows by antisymmetrically expanding the transformed creation operators, and retains the fermionic ordering signs.

In particular, an ordered k-particle word in one copy and vacuum in the other returns with amplitude u^k. Complete transfer has amplitude v^k, or $(-v)^k$ in the reverse direction. The k used below is even, so these transfer amplitudes are positive. The summed occupation of each physical mode across the two copies is conserved; different occupation words cannot contribute to the same return or complete-transfer matrix element.

B. Exact contraction

For one resource block, the replicated vacuum and fully occupied terms each return with amplitude one. The other two replicated terms give return and transfer amplitudes u^4 and v^4. Consequently,

$$\begin{aligned}f(\alpha) := F_4(C;\alpha) &= \frac{1}{2} + \frac{1}{2}(u^4 + v^4)\\ &= 1 - \frac{1}{4}\sin^2(2\alpha).\end{aligned} \tag{13}$$

Thus $3/4 \leq f(\alpha) < 1$ on the interior domain, and $f(\pi/4) = 3/4$.

In the doubled state, A_mA_m, B_mB_m, and the span of A_mB_m, B_mA_m have respectively zero, two, and one particles across the first-flag copies. Conservation separates these sectors. Their contributions are p^2, $q^2 f(\alpha)^m$, and a cross term. Local factorization introduces no extra signs: in moving from register-first to block-first Fock order, each crossing interchanges two even local occupation words. Its sign is therefore $+1$.

The total weight of components of B_m with r occupied four-mode blocks is $2^{-m}\binom{m}{r}$. Each such occupation word has $k = 2 + 4r$ particles. The return and complete-transfer contributions give

$$\begin{aligned}K_m(\alpha) &= 2^{-m} \sum_{r=0}^{m} \binom{m}{r} \left(u^{2+4r} + v^{2+4r}\right)\\ &= u^2 \left(\frac{1+u^4}{2}\right)^m + v^2 \left(\frac{1+v^4}{2}\right)^m.\end{aligned} \tag{14}$$

The full parent overlap is therefore

$$\boxed{F_{4m+2}(\psi_{m,q};\alpha) = p^2 + q^2 f(\alpha)^m + 2pqK_m(\alpha).} \tag{15}$$

This is an exact finite-mode identity for all stated m, q, α, with no omitted channels or large-system approximation. The remaining number-state flag in each measured branch is Gaussian. Its crossings with the even resource blocks also have positive sign, including for the odd output ϕ_1. Hence

$$F_{4m+1}(\phi_0;\alpha) = 1, \qquad F_{4m+1}(\phi_1;\alpha) = f(\alpha)^m. \tag{16}$$

Keeping the measured mode as its recorded Gaussian number state would give the same entropy.

IV. EXACT FINITE VIOLATION AND AMPLIFICATION

For the recorded measurement, define the average increase

$$\begin{aligned}\Delta_{m,q}(\alpha) &:= \sum_b p_b P_{4m+1}(\phi_b;\alpha) - P_{4m+2}(\psi_{m,q};\alpha)\\ &= \ln F_{4m+2}(\psi_{m,q};\alpha) - qm\ln f(\alpha). \end{aligned} \tag{17}$$

Strong monotonicity would require $\Delta_{m,q} \le 0$. At $\alpha = \pi/4$, Eqs. (13) and (14) simplify to

$$f = \frac{3}{4}, \qquad K_m = \left(\frac{5}{8}\right)^m. \tag{18}$$

Choose $m = 8$, $q = 1/4$, and $p = 3/4$. The 34-mode input is

$$|\psi\rangle = \frac{\sqrt{3}}{2}|00\rangle|0000\rangle^{\otimes 8} + \frac{1}{2}|11\rangle\left(\frac{|0000\rangle + |1111\rangle}{\sqrt{2}}\right)^{\otimes 8}. \tag{19}$$

Its exact overlaps are

$$\begin{aligned} F_{\rm in} &= \frac{9}{16} + \frac{1}{16}\left(\frac{3}{4}\right)^8 + \frac{3}{8}\left(\frac{5}{8}\right)^8 \\ &= \frac{77509155}{134217728}, \end{aligned} \tag{20}$$

$$(F_0, F_1) = \left(1, \frac{6561}{65536}\right). \tag{21}$$

With both probabilities $3/4, 1/4$ included, the mean output entropy is $2\ln(4/3)$. Therefore

$$\Delta_{8,1/4}(\pi/4) = \ln\left(\frac{25836385}{25165824}\right) > 0. \tag{22}$$

Strictness is the integer comparison $25836385 > 25165824 > 0$, rather than a floating-point test. Integrating $1/(1+x)$ gives the rational enclosure

$$\frac{670561}{25836385} < \Delta_{8,1/4}(\pi/4) < \frac{670561}{25165824}. \tag{23}$$

The approximate value 0.0262968862 is only an illustration. This one finite input and one allowed measurement disprove the universal strong-monotonicity inequality at the specified angle. No extension to more complicated protocols is needed. The witness is not asserted to have the smallest possible mode count.

The same construction applies at every fixed $0 < \alpha < \pi/2$ and $0 < q < 1$. Since $K_m \ge 0$, Eq. (15) implies

$$F_{\rm in} \ge p^2, \qquad P_{\rm in} \le -2\ln p, \tag{24}$$

$$\Delta_{m,q}(\alpha) \ge qm[-\ln f(\alpha)] + 2\ln p. \tag{25}$$

The coefficient of m is strictly positive. In particular, every integer satisfying

$$m > \frac{-2\ln p}{q[-\ln f(\alpha)]} \tag{26}$$

gives a strict finite counterexample. For the parameters of Eq. (19), the right-hand side equals eight. The additional positive terms in Eq. (20) make the violation strict already at this equality.

For fixed interior α, each of $(1+u^4)/2$, $(1+v^4)/2$, and $f(\alpha)$ is less than one. Consequently, as the finite integer m increases,

$$F_{\rm in} \longrightarrow p^2, \qquad P_{\rm in} \longrightarrow -2\ln p, \tag{27}$$

while the mean output entropy is exactly $qm[-\ln f(\alpha)]$. Thus the average increase is unbounded along this sequence of finite-dimensional states, as illustrated in Fig. 1. No state in an infinite Fock space is assumed.

The sufficient threshold is not uniform near either angle endpoint: $f(\alpha) \to 1$ as $\alpha \to 0$ or $\pi/2$. At the endpoints the family has unit overlap. The growing-mode result therefore does not contradict a fixed-mode small-angle covariance expansion or its associated strong monotones [1, 6].

V. VERIFICATION AND OPERATIONAL INTERPRETATION

The analytic contraction has two complementary representations. Appendix A derives the complete positive Hermitian-Majorana expansion directly from the graded rotation. Appendix B gives the exact local occupied-minor kernels. Complete local Majorana-string sums reproduce these polynomial identities and the rational certificate. An independently programmed exterior-power calculation also reproduces the same kernels and finite witness. Full graded two-register calculations at $N = 6, 10$ provide additional floating-point checks of copy conventions and pure branches. These checks do not replace the all-m factorization or establish strictness through a tolerance. Exact arithmetic establishes Eq. (22).

OpenAI Codex language-model assistance was used in the state-family search, derivations, code development, literature comparison, and preparation of this draft. The native tool was OpenAI Codex CLI, version 0.162.1. Recorded invocations requested model identifiers `gpt-6.1-sol` with max reasoning and `gpt-5.6-sol` with ultra reasoning; these identify local invocation routing, rather than independently certified service weights. Verification used explicit Majorana-string and occupied-minor computations, including an independent computational reconstruction. Such computational checks are distinct from human scientific review, which remains pending for this local draft.

The limitation concerns using replica twist entropy to bound average pure-state Gaussian conversion yield. An

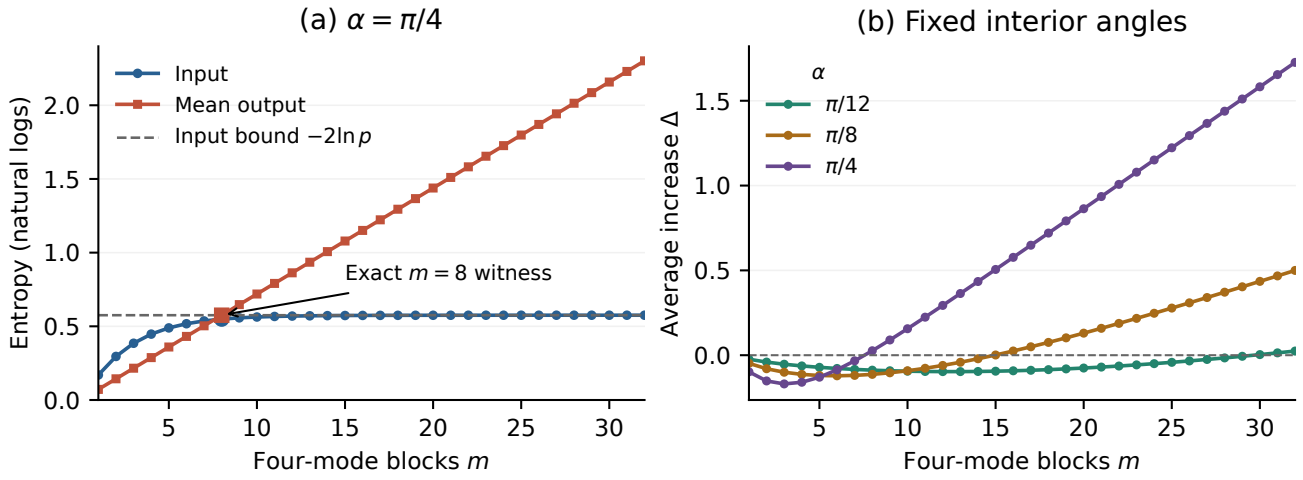


FIG. 1. Recorded occupation measurement of the family in Eq. (8), with $q = 1/4$. (a) At $\alpha = \pi/4$, the input entropy stays below $-2\ln(3/4)$, while the mean output entropy is $m\ln(4/3)/4$. (b) The average increase from Eq. (17) at three fixed interior angles. Markers denote integer m; connecting lines guide the eye. Logarithms in the plot are evaluated in double precision. The $m = 8$ certificate is exact and does not rely on the plot. Plotting code was drafted with OpenAI Codex assistance; all values are generated from the displayed analytic expressions.

occupation measurement reveals an extensive resource in one branch, with both outcomes weighted by their actual probabilities. Meanwhile the coherent parent retains a nonzero p^2 contribution to its two-copy overlap, bounding its logarithm. This explains the failure without introducing a parity-coherent input or discarding an unfavorable outcome.

Covariance-based strong monotones and Williamson-spectrum majorization [1, 2] constrain different functionals. The finite-angle replica quantity depends on higher-order correlations [6]. Likewise, sharing a graded rotation with fermionic convolution does not identify a direct overlap amplitude with the entropy of a reduced convolution output [3, 4]. The bridge spectral framework is another established two-copy representation; monotonicity of its spectral-support degree is a different assertion from averaged monotonicity of a logarithmic finite-angle amplitude [5].

The coherent-dilution strategy is credited to its stabilizer predecessors [7, 8]. The resource block C is a four-qubit stabilizer state, for example with generators $XXXX, ZZII, IZZI, IIZZ$, yet it has fermionic overlap 3/4 at the symmetric angle. Thus the free theories and the functions being tested are distinct. The substantive fermionic calculation is the parity-legal family and its graded all-angle overlap, rather than a new generic dilution mechanism.

The result leaves Gaussian invariance, product additivity, faithfulness, and the source's SYK dynamical analysis intact. It does not establish failure under deterministic pure-to-pure conversion, a smallest counterexample, or an arbitrary mixed-state extension. Its operational conclusion is that the given finite-angle entropy fails Eq. (5), even for a single recorded occupation measurement.

ACKNOWLEDGMENTS

OpenAI Codex also assisted with manuscript drafting and preparation of the reproducibility materials. Author review and confirmation of the draft author details are pending.

DATA AVAILABILITY

All central identities and the strict rational certificate are given in the article. The accompanying local reproducibility materials contain the exact Majorana-string checker, the independent occupied-minor checker, the graded finite-mode replay, and the figure-generation code with its data and software requirements.

Appendix A: Positive string expansion with the graded convention

Set $A_j = \gamma_j^{(1)}\gamma_j^{(2)}$. Then $A_j^2 = -I$, and $A_jA_k = A_kA_j$ for $j \neq k$. With $c = \cos(\alpha/2)$, $s = \sin(\alpha/2)$,

$$U_{\mathrm{R}}(\alpha) = \prod_{j=1}^{2N}(cI + sA_j). \tag{A1}$$

For an ordered subset S of length k, let $\gamma_S = \prod_{j\in S}\gamma_j$ and $h = k(k-1)/2$. The ordinary register representation gives

$$\prod_{j\in S} A_j = (-1)^h \gamma_S \Pi_N^k \otimes \gamma_S. \tag{A2}$$

Odd-k expectations vanish in a parity eigenstate. For even k, define the Hermitian string $\chi_S = i^h \gamma_S$. Its expectation is real, and the expectation of Eq. (A2) in the replicated input equals $|\langle\psi|\chi_S|\psi\rangle|^2$. Expanding Eq. (A1) yields the published expression [6]

$$F_N(\psi;\alpha) = \sum_{\substack{S\subseteq\{1,\dots,2N\}\\ |S| \text{ even}}} c^{2N-|S|} s^{|S|} |\langle\psi|\chi_S|\psi\rangle|^2. \quad \text{(A3)}$$

All weights are positive on the stated domain, and the identity term is strictly positive. Unitarity supplies the upper bound one. This derivation also specifies the string phase and the replica angle independently of a tensor-copy commutation shortcut.

Appendix B: Complete surviving local kernels

For local vectors x_0, x_1 define a replica kernel

$$T_{abcd} = \langle x_a|\langle x_c|U_{\mathrm{R}}(\alpha)|x_b\rangle|x_d\rangle. \quad \text{(B1)}$$

Use $x_0 = |00\rangle, x_1 = |11\rangle$ for the flag kernel T^{fl}, and $x_0 = V, x_1 = C$ for the resource kernel T^{res}. The occupied minors of Eq. (12) give Table I, where $t = \sin^2\alpha$, $a = [1+(1-t)^2]/2$, and $b = (1+t^2)/2$. All ten omitted flag entries vanish: particle number in each mode pair must match, so the flag occupations can only return or transfer together. Resource entries paired with a zero flag entry cannot contribute.

For example, the 0011 resource element averages the vacuum return and the four-particle return, giving $(1+u^4)/2$. The 0110 element averages the vacuum and complete transfer, giving $(1+v^4)/2$. The 1111 element is Eq. (13). The two-flag return and transfer are respectively $u^2 = 1-t$ and $v^2 = t$.

Writing $z_0 = \sqrt{p}$, $z_1 = \sqrt{q}$, contraction of the four parent coefficients gives

$$\begin{aligned} F_{\mathrm{in}} &= \sum_{a,b,c,d=0}^{1} z_a z_b z_c z_d T^{\mathrm{fl}}_{abcd} \big(T^{\mathrm{res}}_{abcd}\big)^m \\ &= p^2 + q^2(1-t+t^2)^m + 2pq\big[(1-t)a^m + tb^m\big]. \end{aligned} \quad \text{(B2)}$$

The block ordering signs are positive because every local occupation word in both bra and ket has even parity.

TABLE I. The six surviving local replica kernels. Every value is an exact polynomial in $t = \sin^2\alpha$.

$abcd$	T^{fl}_{abcd}	T^{res}_{abcd}
0000	1	1
1111	1	$1-t+t^2$
0011, 1100	$1-t$	$[1+(1-t)^2]/2$
0110, 1001	t	$(1+t^2)/2$

This proves factorization for every integer m, rather than inferring it from finite tests. Equation (B2) is exactly Eq. (15).

[1] L. Leone and L. Bittel, Fermionic entropy: an efficiently measurable strong monotone for non-gaussianity (2026), arXiv:2607.29670v1 [quant-ph].

[2] X. Turkeshi, P. Sierant, and P. S. Tarabunga, Williamson majorization theory of fermionic non-gaussianity (2026), arXiv:2608.10140v2 [quant-ph].

[3] X. Lyu and K. Bu, Fermionic Gaussian testing and non-gaussian measures via convolution (2024), arXiv:2409.08180v2 [quant-ph].

[4] L. Coffman, G. Smith, and X. Gao, Measuring non-gaussian magic in fermions: Convolution, entropy, and the violation of Wick's theorem and the matchgate identity (2025), arXiv:2501.06179v1 [quant-ph].

[5] P. S. Tarabunga, Fermionic non-gaussianity via Bell sampling: monotones and efficient quantum algorithms (2026), arXiv:2606.05066v2 [quant-ph].

[6] N. Sun and P. Zhang, Characterizing fermionic non-gaussianity in the Sachdev–Ye–Kitaev model via replica twist entropy (2026), arXiv:2610.10123v1 [quant-ph].

[7] T. Haug and L. Piroli, Stabilizer entropies and non-stabilizerness monotones, Quantum **7**, 1092 (2023), arXiv:2303.10152v2 [quant-ph].

[8] L. Leone and L. Bittel, Stabilizer entropies are monotones for magic-state resource theory, Phys. Rev. A **110**, L040403 (2024), arXiv:2404.11652v3 [quant-ph].